

Unsigned Difference Norms in the Banach Function Space Case

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Source and Target

The source paper is Nick Lindemulder, *Difference Norms for Vector-Valued Bessel Potential Spaces with an Application to Pointwise Multipliers*, arXiv:1505.08039. Corollary 4.9 proves a square-function difference norm characterization for $H_p^s(\mathbb{R}^d, w; X)$ when X is a UMD Banach function space. Immediately after it, Remark 4.10 asks whether one may replace the signed averaged differences $K_m(2^{-j}, f)$ in the right hand side by unsigned difference averages $d_K^m(2^{-j}, f)$.

characterization for $F_{p,q}^s(\mathbb{R}^d; X)$.

4.4. The Special Case of a Banach Function Space. In the special case that X is a Banach function space, we obtain the following corollary from the main result Theorem 4.1:

Corollary 4.9. *Let X be a UMD Banach function space, $s > 0$, $p \in (1, \infty)$, $w \in A_p(\mathbb{R}^d)$ and $m \in \mathbb{N}$, $m > s$. Suppose that $K \in \mathcal{K}(\mathbb{R}^d) \cap L^1(\mathbb{R}^d, (1 + |\cdot|)^{(d+3)m})$ satisfies the Tauberian condition (26) for some $c, \epsilon > 0$. For all $f \in L^p(\mathbb{R}^d, w; X)$ we then have the equivalence of extended norms*

$$(35) \quad \|f\|_{H_p^s(\mathbb{R}^d, w; X)} \approx \|f\|_{L^p(\mathbb{R}^d, w; X)} + \left\| \left(\sum_{j=1}^{\infty} |2^{js} K_m(2^{-j}, f)|^2 \right)^{1/2} \right\|_{L^p(\mathbb{R}^d, w; X)}.$$

Proof. By the Khintchine-Maurey theorem, the right-hand side (RHS) of (35) defines an extended norm on $L^p(\mathbb{R}^d, w; X)$ which is equivalent to $\|\cdot\|_{H_p^s(\mathbb{R}^d, w; X)}^{(m,K)}$. Therefore, we only need to check the $\mathcal{R}$ -boundedness condition in Theorem 4.1(ii). But this follows from Proposition 4.11 below (and the discussion after it). $\square$

Remark 4.10. Let X be a UMD Banach function space, $s > 0$, $p \in (1, \infty)$, $w \in A_p(\mathbb{R}^d)$ and $m \in \mathbb{N}$, $m > s$. Suppose $K \in \mathcal{K}(\mathbb{R}^d)^+ \setminus \{0\}$. Then it is a natural question whether we can replace $K_m(2^{-j}, f)$ by $d_K^m(2^{-j}, f)$ in the RHS of (35), where

$$d_K^m(t, f)(x) := \int_{\mathbb{R}^d} K(h) |\Delta_h^m f(x)| dh, \quad t > 0, x \in \mathbb{R}^d.$$

In view of the domination $|K_m(t, f)| \leq d_K^m(t, f)$, this is certainly true for the inequality ' $\lesssim$ ' in (35). For the reverse inequality ' $\gtrsim$ ' one could try to extend the maximal function techniques from [53, Proposition 6] to our setting via the square function variant of the Littlewood-Paley characterization (13); here one would have to replace the classical Hardy-Littlewood maximal function by the Banach lattice version from [4, 12, 50].

Proposition 4.11. *Let X be a UMD Banach function space, $p \in (1, \infty)$ and $w \in A_p(\mathbb{R}^d)$.*

There is an evident typo in the displayed definition in Remark 4.10: the scale t is missing from the difference. The surrounding sentence $|K_m(t, f)| \leq d_K^m(t, f)$ only makes sense for

$$D_K^m(t, f)(x) := \int_{\mathbb{R}^d} K(h) |\Delta_{th}^m f(x)| dh.$$

The theorem below answers the question for this intended scaled definition, under the same standing hypotheses as Corollary 4.9.

Proof Intuition

The signed square function in Corollary 4.9 is controlled by the source paper through a Littlewood-Paley decomposition $f = \sum S_n f$. The unsigned quantity is sublinear rather than linear, but in a Banach function space the lattice modulus and the ℓ^2 -square function let us keep the same decomposition. High-frequency pieces are controlled by the source ℓ^2 -boundedness of positive convolution kernels. Low-frequency pieces are controlled exactly as in the proof of Theorem 4.1(ii): for each fixed h , the difference $\Delta_{2^{-j}h}^m$ is represented by the multiplier family of Lemma 4.8, and the integrability assumption on K absorbs the factor $(1 + |h|)^{(d+3)m}$. The source inequality $|K_m(t, f)| \leq D_K^m(t, f)$ gives the converse side.

Theorem 1. *Let X be a UMD Banach function space, let $s > 0$, $p \in (1, \infty)$, $w \in A_p(\mathbb{R}^d)$, and let $m \in \mathbb{N}$ with $m > s$. Assume that*

$$K \in \mathcal{K}(\mathbb{R}^d)^+ \cap L^1(\mathbb{R}^d, (1 + |\cdot|)^{(d+3)m})$$

and that K^{Δ^m} satisfies the Tauberian condition of Lindemulder's Theorem 4.1(i), equivalently the condition used in Corollary 4.9. Then, for every $f \in L^p(\mathbb{R}^d, w; X)$,

$$\|f\|_{H_p^s(\mathbb{R}^d, w; X)} \simeq \|f\|_{L^p(\mathbb{R}^d, w; X)} + \left\| \left(\sum_{j=1}^{\infty} |2^{js} D_K^m(2^{-j}, f)|^2 \right)^{1/2} \right\|_{L^p(\mathbb{R}^d, w; X)}.$$

The constants depend only on the parameters in the source corollary and on $\int K(h)(1+|h|)^{(d+3)m} dh$.

Proof. Write

$$U_J(f) := \left\| \left(\sum_{j=1}^J |2^{js} D_K^m(2^{-j}, f)|^2 \right)^{1/2} \right\|_{L^p(\mathbb{R}^d, w; X)}.$$

It is enough to prove $U_J(f) \lesssim \|f\|_{H_p^s(\mathbb{R}^d, w; X)}$ uniformly in J , since the opposite estimate follows from Corollary 4.9 of the source paper and the lattice inequality

$$|K_m(t, f)| = \left| \int_{\mathbb{R}^d} K(h) \Delta_{th}^m f dh \right| \leq \int_{\mathbb{R}^d} K(h) |\Delta_{th}^m f| dh = D_K^m(t, f).$$

Fix a Littlewood-Paley decomposition $(S_n)_{n \geq 0}$ as in the source paper and put $f_n = S_n f$ for $n \geq 0$, $f_n = 0$ for $n < 0$. For each $j \geq 1$, $f = \sum_{n \in \mathbb{Z}} f_{n+j}$ in the relevant distribution sense. Since $D_K^m(t, \cdot)$ is subadditive and positive, the lattice Minkowski inequality gives

$$U_J(f) \leq \sum_{n \in \mathbb{Z}} \left\| \left(\sum_{j=1}^J |2^{js} D_K^m(2^{-j}, f_{n+j})|^2 \right)^{1/2} \right\|_{L^p(\mathbb{R}^d, w; X)}.$$

First let $n > 0$. From

$$|\Delta_{th}^m g| \leq \sum_{\ell=0}^m \binom{m}{\ell} |g(\cdot + (m-\ell)th)|$$

we get

$$D_K^m(t, g) \leq \sum_{\ell=0}^m \binom{m}{\ell} P_{\ell, t} |g|,$$

where $P_{\ell,t}$ is either multiplication by $\int K$ or convolution with a positive dilation/reflection of K . These kernels form a bounded set in $\mathcal{K}(\mathbb{R}^d)$. Proposition 4.11 of the source paper says that such bounded sets of convolution kernels are ℓ^2 -bounded on $L^p(\mathbb{R}^d, w; X)$ when X is a UMD Banach function space. Hence

$$\begin{aligned} & \left\| \left(\sum_{j=1}^J |2^{js} D_K^m(2^{-j}, f_{n+j})|^2 \right)^{1/2} \right\|_{L^p(w; X)} \\ & \lesssim 2^{-ns} \left\| \left(\sum_{j=1}^J |2^{(n+j)s} f_{n+j}|^2 \right)^{1/2} \right\|_{L^p(w; X)} \lesssim 2^{-ns} \|f\|_{H_p^s(\mathbb{R}^d, w; X)}. \end{aligned}$$

The last step is the Khintchine-Maurey theorem together with the randomized Littlewood-Paley equivalence (13) in the source paper.

Now let $n \leq 0$. By lattice Minkowski and Khintchine-Maurey,

$$\begin{aligned} & \left\| \left(\sum_{j=1}^J |2^{js} D_K^m(2^{-j}, f_{n+j})|^2 \right)^{1/2} \right\|_{L^p(w; X)} \\ & \leq \int_{\mathbb{R}^d} K(h) \left\| \left(\sum_{j=1}^J |2^{js} \Delta_{2^{-j}h}^m f_{n+j}|^2 \right)^{1/2} \right\|_{L^p(w; X)} dh \\ & \simeq \int_{\mathbb{R}^d} K(h) \left\| \sum_{j=1}^J \varepsilon_j 2^{js} \Delta_{2^{-j}h}^m f_{n+j} \right\|_{L^2(\Omega; L^p(w; X))} dh. \end{aligned}$$

The proof of Theorem 4.1(ii) in the source paper applies Lemma 4.8 to the last Rademacher norm and yields, for every fixed h ,

$$\left\| \sum_{j=1}^J \varepsilon_j 2^{js} \Delta_{2^{-j}h}^m f_{n+j} \right\|_{L^2(\Omega; L^p(w; X))} \lesssim 2^{n(m-s)} (1 + |h|)^{(d+3)m} \|f\|_{H_p^s(\mathbb{R}^d, w; X)}.$$

Integrating in h is allowed by the assumed moment condition on K , so

$$\left\| \left(\sum_{j=1}^J |2^{js} D_K^m(2^{-j}, f_{n+j})|^2 \right)^{1/2} \right\|_{L^p(w; X)} \lesssim 2^{n(m-s)} \|f\|_{H_p^s(\mathbb{R}^d, w; X)}.$$

Summing the estimates over $n > 0$ and $n \leq 0$ gives two convergent geometric series, since $s > 0$ and $m - s > 0$. Thus

$$U_J(f) \lesssim \|f\|_{H_p^s(\mathbb{R}^d, w; X)}$$

uniformly in J . Letting $J \rightarrow \infty$ proves the desired upper estimate by monotone convergence in the Banach lattice $L^p(\mathbb{R}^d, w; X)$. Together with the already proved lower estimate this gives the asserted norm equivalence. $\square$

Verification and Novelty Notes

No computation is used. The proof reduces the open estimate to three source ingredients: Corollary 4.9, Proposition 4.11, and the fixed-step multiplier estimate in Lemma 4.8 as used in the proof of Theorem 4.1(ii).

Bounded duplicate and novelty checks were performed in the local run indexes for arXiv id 1505.08039, the phrase `R-bdd_averaging_operators`, averaging operators, and unsigned/difference-modulus keywords. No existing packet for this question was found. A bounded web search on June 28, 2026 for exact phrases from Remark 4.10 and for `d_K`, `K_m`, and the paper title did not identify a later answer. Novelty confidence is therefore `bounded-search likely new`, not a complete bibliographic audit.

Limitations and Review Recommendation

This packet answers the intended scaled version of Remark 4.10. Read literally, the source display of $d_K^m(t, f)$ omits t from Δ_{th}^m ; that literal formula cannot be the intended one because it would not imply the displayed domination $|K_m(t, f)| \leq d_K^m(t, f)$.

Human review should focus on the high-frequency use of Proposition 4.11 for the positive convolution majorants and on the transfer of the source low-frequency Rademacher estimate through Khintchine-Maurey and lattice Minkowski.

References

- [1] N. Lindemulder, *Difference Norms for Vector-Valued Bessel Potential Spaces with an Application to Pointwise Multipliers*, arXiv:1505.08039.